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Lesson Notes

MODULE 5 | LESSON 2

VALUATION CHALLENGES: MARKET FRICTIONS AND MODEL RISK

Reading Time	75 minutes
Prior Knowledge	Subprime mortgage, Option (put and call) pay-off
Keywords	Collateral pool, First and second lien, Moral hazard, Refinance, Cash-out refinance, Market friction, Model risk, Loss Given Default (LGD), Probability of Default (PD), Recovery Rate (RR)

In the previous lesson, we learned the fundamental 5 Cs of credit analysis and applied these to various examples of credit, including MBSs.

In this lesson, we take this kind of analysis further by applying the 5 Cs to a mortgage pool underlying an MBS. We also learn why the structure of the market can make it hard to get reliable credit information, even from a rating agency. We discuss credit concepts such as probability of default and loss given default, which can be used to model the value of MBSs. Given the complexity of MBS modeling, we need to rely more on rating agencies and other market participants, and we ponder government responses to some of the frictions in the market: we look closely at regulations that came after the subprime mortgage crisis and their effectiveness.

1. The 5 Cs Return

The required reading "[Understanding the Securitization of Subprime Mortgage Credit](#)" lists five credit characteristics of the

subprime mortgagor, numbered below for easy reference (Ashcraft & Schuermann 14-16):

1. Two or more 30-day delinquencies in the last 12 months, or one or more 60-day delinquencies in the last 24 months;
2. Judgment, foreclosure, repossession, or charge-off in the prior 24 months;
3. Bankruptcy in the last 5 years;
4. Relatively high default probability as evidenced by, for example, a credit bureau risk score (FICO) of 660¹ or below (depending on the product/collateral), or other bureau or proprietary scores with an equivalent default probability likelihood; and/or,
5. Debt service-to-income ratio of 50 percent or greater; or, otherwise limited ability to cover family living expenses after deducting total debt-service requirements from monthly income.

Recall the 5 Cs of credit analysis and consider how you might use them to evaluate these characteristics.

The first four all suggest a deficiency in the borrower's *character* or *capacity*. Multiple delinquencies (late payments) in the past year or two most likely means the borrower was unwilling or unable to pay on time. Same with bankruptcy and "judgment, foreclosure, repossession, or charge-off," which are actions taken when a borrower never repaid their debt, as opposed to just repaying late. As discussed in the last lesson, a bad (low) credit score is also interpreted as a *character* issue. Lastly, we recognize the "debt service-to-income ratio" as the DTI ratio we discussed in the last lesson, which is another measure of *capacity*.

In "the motivating example" of the same section, the mortgage pool underlying a particular MBS offering is summarized by, among other data, the following (Ashcraft & Schuermann 14-15):

- 98.7% of the mortgage loans are first-lien. The rest are second-lien home equity loans.
- 43.3% are purchase loans, meaning that the mortgagor's stated purpose for the loan was to purchase a property. The remaining loans' stated purpose are cash-out refinances of existing mortgage loans.
- 90.7% of the mortgagors claim to occupy the property as their primary residence. The remaining mortgagors claim to be investors or purchasing second homes.
- The **combined loan-to-value ratio (CLTV)** is the sum of the original principal balance of all loans secured by the property to

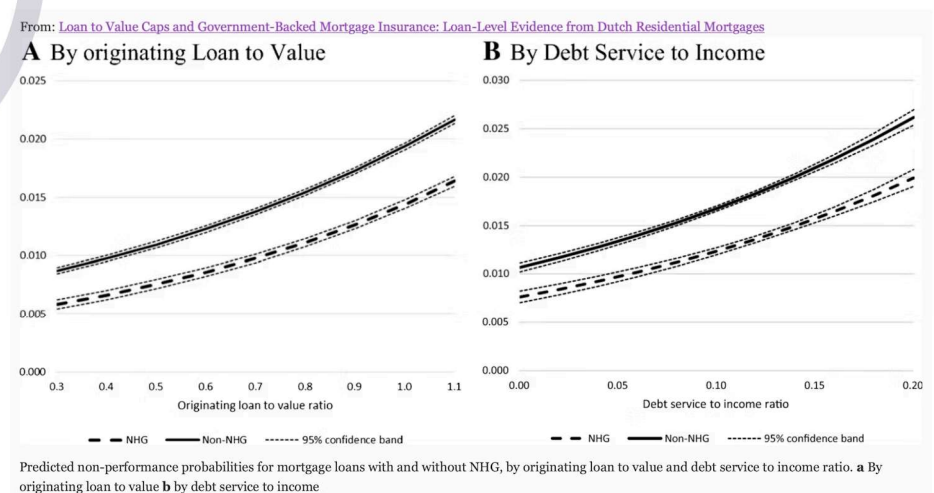
its appraised value. The average mortgage loan in the pool has a CLTV of 80.34%, which implies an average downpayment of 20%. However, given that this is an average number, the astute credit risk analyst should pay attention to the proportion of borrowers with more than 80% CLTV.

Let's take a look at some other data relating to LTV and DTI in the charts below. Observe how the mortgage default rates ("non-performance probabilities") are highly correlated with LTV ratio and credit score.

Two notes about these graphs: 1. The description refers to these as "*predicted* non-performance probabilities," but they are not so much predictions or forecasts as they are smooth lines fit to the actual (empirical) data. 2. "NHG" stands for "Nationale Hypotheek Garantie," which is insurance provided by a Dutch government-owned fund, in many ways similar to Fannie Mae or Freddie Mac in the United States or the Japan Housing Finance Agency. The distinction between NHG and non-NHG is not relevant for our current discussion; what is relevant is the upward slope in both graphs representing significant positive correlation between LTV and default and between DTI ("Debt service to income" below) and default.

As you can see, a high LTV, especially as it approaches 100 (equivalent to the loan amount equal to the house value, meaning no down payment), is bad because it suggests a higher probability of default. As you move from left to right (i.e., the LTV increases), the PD increases. Very similarly, the probability of default increases as DTI increases.

Figure 1: Default Rates by LTV and DTI



Source: de Haan, Leo, and Mauro Mastrogiacomio. "Loan to Value Caps and Government-Backed Mortgage Insurance: Loan-Level Evidence from Dutch Residential

Mortgages." *De Economist*, vol. 168, 2020, p. 466. <https://doi.org/10.1007/s10645-020-09367-w>.

A first-lien loan has priority over a second-lien loan. (From the perspective of the five Cs, the implications of this difference might be captured under "Collateral"; alternatively, they might be classified under "Conditions" because whether the loan has a first- or second-lien is a "contractual term of the loan.") If a borrower has a second-lien loan and the borrower defaults, then the lender of the first-lien loan is repaid from the proceeds of the liquidated collateral first. The lender of the second-lien loan only benefits if the value of the collateral (a house in this case) exceeds the amount of the first-lien loan. As such, for the MBS investor reviewing the mortgage pool referenced above, the fact that almost all of the loans are first-lien is good: this *collateral* is better than a similar pool with a higher proportion of second-lien loans.



The first lien is senior to the second lien, meaning the creditor or lender of the first lien has priority for the value of the collateral: if there is a default, the house will be sold, and the first-lien lender is paid back from the proceeds completely or as much as possible. Once that has happened, the second-lien lender is repaid to the extent that there are any proceeds left after paying the first-lien lender.

In the last lesson, we mentioned that "conditions" can be interpreted in different ways. One condition is the purpose of the loan. Recall that you might prefer to lend to Nushi so she can get to work (to earn money and pay you back) but *not* so that she can spend (the money *you* loaned her) frivolously. In the same way, you might prefer to lend Nushi the money she needs to buy a home: that is, "purchase a property" and "occupy the property as her

primary residence" in the language above. And you are less willing to lend her money so that she has extra cash to spend however she wants.

Not all mortgages help the borrower buy a new house. Sometimes, the borrower already has the house and a mortgage, and they take out a new mortgage to replace the old one. This is called a **refinancing**: it allows the borrower to reset their mortgage terms, often so that they get a better (lower) interest rate. There are different ways to refinance depending on the reason for the refinance. For example, a **cash-out refinance** is for the borrower who has already repaid some of the principal on an existing mortgage. By replacing the existing mortgage, the borrower can effectively withdraw that principal that they had paid and do whatever they want with the cash-out amount. As with Nushi, this may be an undesirable *condition* for an underlying mortgage.

Similarly, you may not want to lend money so that a borrower can buy a second home. Since it's not where they live as their primary residence, the borrower may be more careless about repaying the mortgage. After all, if they lose their second home, at least they still have their first home. By the same token, you may not want to lend money to an investor who is buying a property so that they can resell (flip) it at a higher price. Since the investor already has a home, they may not care as much about the mortgaged property; if something goes wrong with their investment (a natural disaster or a downturn in the economy), the investor could just walk away by making no further mortgage payments and defaulting on the mortgage.

As this relates to the mortgage pool characteristics above, you can compare the percentage of purchase loans (43.3%) to the percentage in another similarly priced MBS mortgage pool to decide which is preferable. And you can do the same with the percentages of primary residence, secondary residence, and investor borrowers. Solid credit analysis of MBSs takes these *conditions* into account.

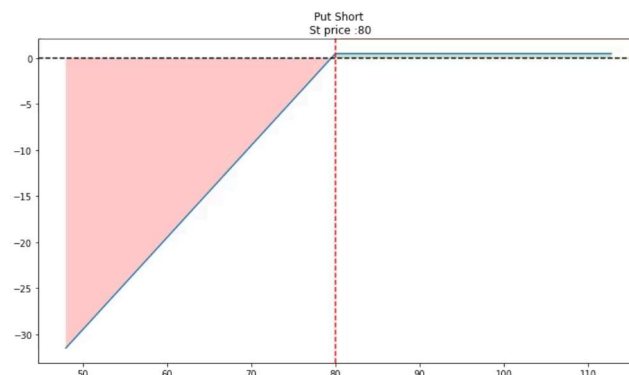
So far, we have related the mortgage pool characteristics to *character, capacity, collateral, and conditions*. That leaves one C unmentioned: *capital*. As mentioned in Ashcraft and Schuermann as well as in the previous lesson's discussion of the mortgage application, the lender is generally interested in the borrower's assets. Just as it was a good sign that Nushi had a nice TV and a nice watch, it is an even better sign if she has a good amount of money in the bank and/or a decently sized securities portfolio, i.e.,

strong net assets. But mortgages are generally non-recourse loans: even though these (net) assets are a good sign, the lender generally does not have recourse to these other assets. That is, the lender can take ownership of the house, which collateralizes the mortgage, but the lender cannot take ownership of any *other* property belonging to the borrower. This makes it even easier for some borrowers, especially the secondary-residence and investor borrowers, to walk away from their mortgages. That means the borrower's *capital* is less relevant than in the context of a recourse loan, but it is still meaningful; otherwise, mortgage lenders (banks) would not require this information.

The non-recourse nature of the mortgage also introduces non-linearity into the profit-and-loss profile of the loan, similar to the non-linearity we saw in the discussion of profit and loss from option trades in Module 4, Lesson 1. The relevant graphs from that section are presented again for you below to conveniently visualize the similarity in the risk and reward between options and mortgages.

Say, for example, a bank provides an \$80,000 loan to a borrower to buy a \$100,000 house (the borrower made a \$20,000 down payment): This is analogous to the bank having a short put with an \$80,000 strike. A few months later, the local real estate market deteriorates, and the house is now worth \$80,000. In this situation, most borrowers will continue to make their mortgage payments and avoid default. Though the house price decreased significantly, the bank has not lost any money—yet.

Figure 3: Short Put Payoff Diagram

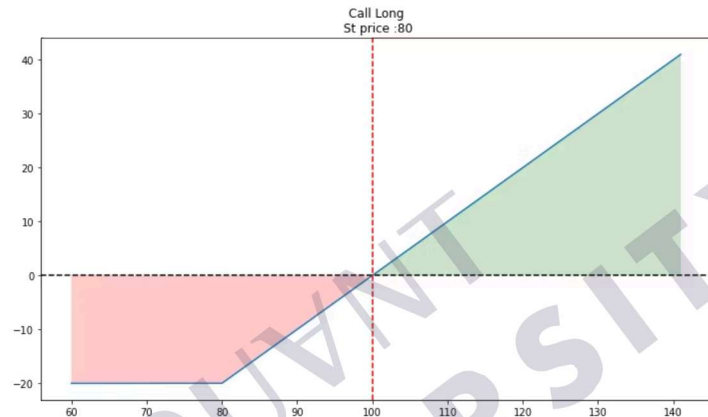


Now, say the real estate market sinks even more, and the house is only worth \$50,000. (Such a loan is called "under water" because the home value is worth less than the outstanding loan amount.) At this point, many mortgage borrowers will decide to walk away, as described above. If the borrower does default, the bank suddenly has a loss of at least \$30,000 (check the graph): the

bank's P/L has moved down the diagonal into a loss. This broken line demonstrates the non-linearity of risk to the bank.

By this same logic, the borrower has a long call position on her house. If her house price increases substantially, she can sell her house, pay back the mortgage loan, and make a profit. Her P/L profile is shown in the graph below: she has all of the upside (potential profit from house price appreciation) and hardly any of the downside (risk of loss).

Figure 4: Long Call Payoff Diagram



2. NINJAs, Liars, and Information Asymmetry

As discussed in the readings, more aggressive, or risky, underwriting allows for more relaxed documentation standards. These loans are also called "no doc" or "low doc" loans, referring to the level of documentation. And here, documentation simply refers to the credit applicant providing proof (documents) of such things as the borrower's income and assets. The borrower may answer the lender's questions about income and assets but doesn't have to document or prove their answers. For example, the borrower for a **SIVA** mortgage has merely **stated** their **income** but **verified** their **assets**. A **SISA** mortgage borrower has merely **stated** their **income** and **stated** their **assets**. These types of loans had been called "liar loans" because of the ease and frequency with which borrowers could exaggerate their *capacity* and/or *capital* in a mortgage application. Lenders of such loans would rely heavily on the borrower's credit score as an indicator of *character* and overall creditworthiness.

For SIVA, SISA, and **NINA** (**no income** and **no assets** information is provided by the borrower) loans, the lender might still at least check that the borrower is in fact employed where they say they are employed. Enter the **NINJA** loan: **No income** and **no job** or **assets** information is provided by the borrower to the lender. The

NINJA borrower is silent on these *capacity* and *capital* characteristics.

3. Information and Moral Hazard

"Informational friction" or "asymmetric information" describes a situation where one party to a transaction or agreement has more or better information than the other. The information is not symmetric across the two parties. For example, the borrower under a "no doc" or "low doc" loan knows how much they have in assets and how much they make in income, but the lender does not have any certainty about these figures.

Asymmetric information easily leads to moral hazard: the party with better information may be tempted to act in bad faith because the consequences are not consequential enough. Recall that the low-doc or no-doc mortgages were called "liar loans." Moral hazard also describes a situation where one party takes excessive risk for their own benefit because they will not bear the full cost of those risks.

The classic example of moral hazard comes from the insurance industry: once a party or person is insured, they may take more risks than before they were insured. If someone has fire insurance, they may not be as careful to put out the fire in their fireplace at night. If someone has insurance against auto theft, they may be more likely to leave their keys in the car or leave their car parked in a bad neighborhood. And similarly, if there is little or no downside to overstating one's income and assets so that person can get a mortgage loan, they will be tempted to do so. After all, if they default on the mortgage because they can't afford to make the payments, the mortgage is non-recourse, so they can just walk away. A person might also choose to walk away if real estate prices decline significantly: why should someone keep making mortgage payments on a principal amount of \$400,000 if the house is now only worth \$300,000? This is a moral hazard and only one of the several moral hazards that beset the financial industry in general and the mortgage and MBS markets in particular.

The reading "Understanding the Securitization of Subprime Mortgage Credit" (Ashcraft and Schuermann) discusses several **informational frictions** and moral hazards in the mortgage and MBS markets and how these created the conditions for the credit crisis in the United States. Pay close attention to how these dynamics played out. Given the profit motive of most financial

players, it pays to become very familiar with how incentives affect the behavior of other participants in whichever market you are active.

4. The Hand That Feeds: Conflicts of Interest and Rating Agencies

For example, the reading also describes how certain incentive structures created a special type of moral hazard called "conflict of interest" among various market participants. A conflict of interest arises when a party has more than one interest, and these interests are in conflict. Recall that Nushi is looking for a 4,000 Nigerian Naira loan from you. Say you don't know Nushi well, so you don't know her *character*, i.e., whether you can trust her. So you would be willing to lend her the cash against her nice-looking watch as *collateral*, but only if you know the watch is worth more than the amount you are lending. Who should you ask to find out the value of Nushi's watch? It would be easy to ask Nushi since it's her watch, but she has a conflict of interest, that is, two interests that conflict. Her primary interest may be getting the loan from you. Telling you the truth about how much she paid for the watch (or what condition it's in, whether it works, etc.) may be only a secondary interest. If Nushi prioritizes her interest in getting a loan over her interest in telling the truth, you as the lender will only have yourself to blame because you did not take into account Nushi's conflict of interest. Better to avoid the moral hazard of tempting Nushi to exaggerate (lie) about the collateral value and find another independent source for this information. Independent in this context means finding someone without a conflict of interest: Nushi's boyfriend Obie and Nushi's mother Amara are not independent; they want Nushi to get what she wants and perhaps they even benefit themselves if Nushi can get this loan.

For the same reason, you would not rely on information from Obie or Amara about Nushi's *character* because of their conflicts of interest. And yet, many creditors and other investors rely on information from rating agencies, despite the fact that rating agencies are literally paid by the issuer of the debt they are rating. (The reading "The 2007–2009 Financial Crisis: An Erosion of Ethics" provides details on how all of these improprieties, and more, made that crisis inevitable.) As you read more about the role and reliability of rating agencies in Box 3 of the Banco de Mexico's "Financial System Report 2007," consider to what extent you should feel comfortable leveraging their analyses.

5. Model Risk: "All models are wrong but some are useful."

This aphorism attributed to the British Statistician George Box (1919-2013) may oversimplify things, but it makes the point that no one model is correct in every way. Nevertheless, if we make judicious use of models, considering carefully what they purport to tell us and being careful about the inputs and assumptions, then they are indeed useful.

When there is a possibility of default, it is rare to lose everything, and the amount one can lose also depends on whether the loan/investment is collateralized or not. To understand the necessary concepts, we start by introducing some terminology.

Probability of Default (PD) is a term used to measure the likelihood that a borrower will default on their debt obligations within a specific time frame. It is a critical component in credit risk assessment and is crucial in credit risk modeling and more. PD is typically estimated using statistical models and historical data. Among the models, the most typical are logistic regression and credit scoring models.

Loss given default (LGD) is the estimated amount of money a bank or other financial institution loses when a borrower defaults on a loan, and it is given as a percentage of total exposure at the time of default or, equivalently, as the amount that we could lose for every single dollar value of investment. A financial institution is very likely to have more than one investment, and therefore, its **total LGD** is calculated after a review of all outstanding loans using cumulative losses and exposure. It is a basic ingredient of most risk models, such as the Basel II model.

LGD is used in the calculation of economic capital, expected loss, and regulatory capital. The expected loss is calculated as a loan's LGD multiplied by both its **probability of default (PD)** and the financial institution's **exposure at default (EAD)**.

EAD is the total value of a loan that a bank is exposed to when a borrower defaults. If a borrower takes out a loan for \$1,000,000 and two years later the amount left on the loan is \$700,000, and the borrower defaults, the exposure at default is \$700,000 not \$1,000,000.

The **default recovery rate (RR)** can be described as the percentage of each dollar invested that is going to be recovered in case of a default. As we wrote earlier, it is rare to lose everything. Thus, if we

have invested \$1,000,000 in bonds with a given rating, we may know from historical data that in case of default the recovery rate is 40%. This means that for every \$1 invested, we will be able to recover \$0.40. It seems plausible that the better the rating of the bond, the higher the RR will be.

This leads us to finally begin to describe LGD via at least two formulae:

$$\text{LGD (dollar amount)} = \text{EAD} \times (1 - \text{RR}).$$

and

$$\text{LGD (as percentage)} = 1 - \frac{\text{Potential Sale Proceeds}}{\text{Outstanding Debt}}.$$

The second formula is useful when there are forms of collateral. Note that the RR has to be estimated, so the LGD estimated with the first formula could seem more technical; however, it might actually sell a formula as more mathematical than it is in reality.

Example. A bank has loaned \$400,000 to a firm. The firm receiving the loan has agreed to put up certain nonproducing assets for a value of \$200,000 as collateral. After 2 years, the amount still to be repaid is \$320,000. The bank estimates that for this type of loan, the probability of default is 6%, the recovery rate can be estimated at 50%, and the bank's analysts estimate that the collateral could be sold quickly for \$190,000. What is the LGD for the bank on this particular loan? What is the expected loss?

Answer. If we ignore the collateral, then we could use the first formula, i.e.,

$$\text{LGD (dollar amount)} = 320,000 \times (1 - 50\%) = 160,000.$$

Considering the presence of collateral, instead we would find

$$\text{LGD (as percentage)} = 1 - \frac{190,000}{320,000} = 40.63\%.$$

The expected loss has to employ the probability of default, that is:

$$\text{expected loss (dollar amount)} = \text{PD} \times \text{LGD} = 6\% \times 160,000 = 9,600.$$

Considering collateral, the expected loss can be computed as

$$\text{expected loss (dollar amount)} = 320,000 \times 6\% \times 40.63\% = 7,801.$$

It is often interesting and important to model the relationships between the variables just introduced. These may be highly nonlinear and may change over economic cycles. Hence, you should appreciate George Box's aphorism at this point.

Some investors may question the indication from rating agencies that are supposed to utilize very sophisticated models as the agencies could very well have a conflict of interest: if they give low ratings, the companies using their services may no longer do that, and therefore, exaggerating the ratings would be in the agencies' own interests.

As an investor in MBSs, one way to avoid the rating agencies' conflicts of interest would be to perform your own credit analysis (or have such analysis performed in house, i.e., within your own trustworthy firm). Another benefit of an independent analysis (if done well) could be the minimization of both model risk and the risk of financial loss due to bad decision making as a result of a bad model. Indeed, the main culprit in the subprime mortgage crisis, according to the readings, was the *failure of rating agencies' models to incorporate appropriate assumptions and valid inputs*. This is called model failure, and the risk of such failure is called "model risk." Since rating agencies' models were considered proprietary, investors could not verify their appropriateness and validity. A firm might be tempted to not get rated if they expect to receive a poor rating. However, smart investors see through this and quickly learn to assume that companies or securities with no rating have very low credit quality.

Certainly, constructing a good credit risk model for a huge pool of mortgages with widely varying characteristics (LTV and DTI ratios, borrower credit scores, loan purposes, etc.) is not simple. With that in mind, [read the article "Determination of Default Probability by Loss Given Default"](#) by Mišanková et al. to understand how the most important variables, such as loss given default (LGD) and probability of default (PD), are related. Also, consider carefully which of the inputs to these models can be known exactly and which ones need to be calculated, which may also require a model and thus implies model risk. Remember, no single model provides a complete or perfect interpretation of the world. Models cannot replace judgment that comes with true understanding. When a single investor relies blindly on a model, they will probably incur significant losses; when many investors make this mistake, the problem may become much larger, even systemic.

The math presented should not be difficult. Don't concern yourself too much with the technical discussion of the many variants of the fundamental formulae, though you should expect to learn about most of the regression approaches mentioned in later coursework. We will return to these formulae when we build out our own valuation model for a hypothetical MBS in the next course: Financial Data.

6. Regulation

Given the complexity of MBSs to model and value (which we will appreciate even more in the next course), investors have naturally relied on rating agencies' assessments. Actually, as you know from the readings, many participants in the mortgage and MBS market relied on other participants' truthfulness though they should not have. In markets where investors do not have enough information to make investment decisions in important markets (such as the housing market), the risks to that market may justify government intervention. This can take many forms, but regulation and bailouts are the most pertinent to a discussion of the financial crisis.

The first part of "Erosion" deals with the many causes of the crisis. The section titled "Reluctant Regulators" lists multiple United States authorities that could have done more to prevent the crisis. As you read through the rest of this paper, consider how things may have turned out differently if regulations were in place.

For example, the section "Excessive Leverage" describes how the increased use of various derivatives added another layer of (synthetic) leverage. Could one of the regulators have put a limit on an investment bank's leverage? In fact, they did—*after* the crisis, as you read in the Dodd-Frank part of the reading. And what of the complex derivatives that enabled synthetic leverage? Yes, Dodd-Frank addressed this type of risk as well. As for the rating agencies' conflicts of interest (recall that they were paid by the issuers of the debt they were rating) and their hidden models, could regulators require that rating agencies be transparent about their modeling methodologies, as well as the "track record" (past performance statistics) of those models in predicting default? Yes—again, regulators addressed these issues in Dodd-Frank, as described in the reading.

But can all of the market's weaknesses and risks be simply regulated away? Read the short article ["Dodd-Frank Act: How Financial Reform May Be Going Wrong"](#) (Eisinger and Bernstein), which describes just a few of the very practical challenges that the

post-crisis regulatory reform faced. Some of the practical difficulties described in this article might be overcome with time. Regardless, the debate about the proper scope of government intervention in the markets continues. For a sense of how these types of financial reforms are framed in very political (instead of pragmatic) terms, [read the report "Doomed to Repeat"](#) about the Dodd-Frank legislation as a response to the Great Financial Crisis. It discusses--also from its own very political perspective--the objectives of Dodd-Frank as well as the conservative criticisms of the legislation and provides a retort to these. Ask yourself as you read this if the issues could ever be resolved to both sides' satisfaction. Can the two sides of this argument ever agree?

Regulation, one of this week's themes, applies to every other principle we've encountered thus far. Credit risk, shorting, volatility, correlation, leverage, and nonlinearity all create opportunities for extreme outcomes and require regulation. As we are just scratching the surface of regulation, we will say much more, but for now, think of regulation as a first-order attempt to mitigate some of the extremeness.

Think back to our housing example: consider that income and home prices have some correlation, consider the nonlinearity of home equity, and consider the credit risk that banks have by lending a mortgage. With no money down, these principles combine to show that, if the economy worsens, people can lose their jobs, housing prices might drop, the home could be worth less than the original mortgage, the homeowner/borrower may no longer be able to pay, and the collateral (the home) itself might even be worth less than the loan. Regulation needs to be multifaceted to address the systemic risks in the financial ecosystem.

7. Conclusion

We applied basic and "high-level" credit analysis to a pool of mortgages that underly an MBS. We found that we didn't necessarily have all the credit information that we would like and that such information may not be reliable anyway due to market frictions and moral hazards. We noted that rating agencies face moral hazard in the form of conflicts of interest, and this is just one potential source of model risk. This prompted us to take our own first look at credit risk modeling and the calculation of probability of default. Since modeling risk is hard (and we will return to it in the Financial Data course), we asked ourselves whether moral hazard and other crisis-causing risk factors could

just be regulated away. We learned that, in the United States, regulators attempted to accomplish this *after* the Great Financial Crisis. However, the effectiveness of these efforts are hotly debated along mostly ideological lines.

In the next lesson, we talk about the broader credit market to which MBS belongs. In the credit market, the main risk is credit risk, or the probability of default, so we look at that as well. The credit spread reflects credit risk, so we also talk about the relationship between the credit spread and the probability of default. As we discussed in this lesson, many investors rely on rating agencies for analysis of credit products like bonds. As such, in the next lesson, we also look at the relationship between credit ratings and probability of default levels. Another important risk, in credit markets as well as many other markets, is liquidity risk, so we look at ways to quantify this risk.

Footnotes

1. The main credit scores in the U.S. are numbers in the range of 300-850, which are used to capture the credit quality of individual persons. A higher score suggests a more creditworthy borrower, and a lower credit score a less creditworthy borrower. Credit scores are very similar to credit ratings, which are assigned to companies or securities. One difference is that ratings range from AAA or Aaa (the best credit quality, implying a low probability of default) to C or D (very poor credit quality, implying a high probability of default).

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